

Montserrat Nafria

其他

基本信息

姓名: Montserrat Nafria

所属院校: 巴塞罗那自治大学

地区: Spain

海外经历: 有

学科领域

微电子科学与工程

材料物理

集成电路设计与集成系统

研究领域

微电子器件与制造

薄膜材料与表面物理

纳米电子与先进工艺

微电子封装与可靠性

材料微结构与表征

EDA工具与电路验证

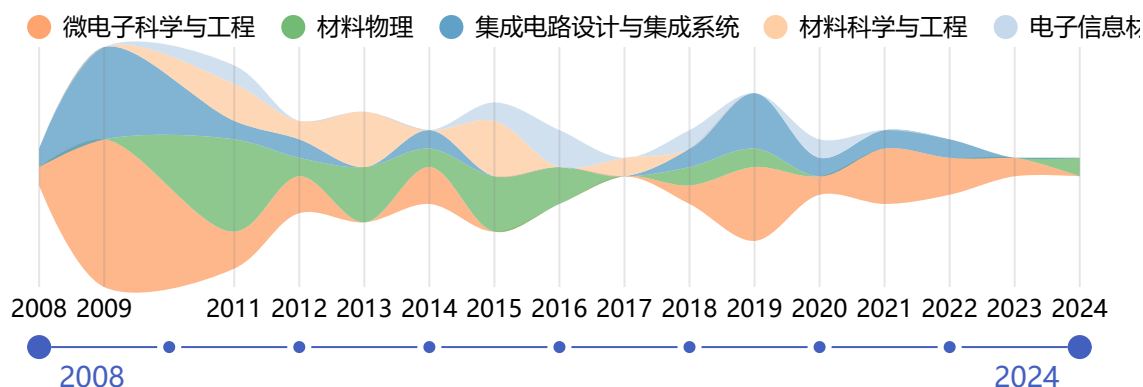
纳米物理与纳米器件

材料缺陷与微结构分析

个人简介

Montserrat Nafria is a teacher at the Autonomous University of Barcelona, Spain. Her D - index in Engineering and Technology is 34, with 5,467 citations, a world ranking of 312, and a national ranking of 190. She is well - known for research in Electron, Graphene, and Scanning tunneling microscope. Her work often integrates fields like NMOS logic, Transistor, and Gate oxide. She frequently publishes in Optoelectronics (74.51%), Electrical engineering (70.59%), and Voltage (64.71%). Some of her highly - cited works include "Soft breakdown conduction in ultrathin (3 - 5 nm) gate dielectrics".

近年发表的论文



On the Aging of OTFTs and Its Impact on PUFs Reliability

MICROMACHINES

Porti, Marc; Palau, Gerard; Crespo-Yepes, Albert; Rus, August Arnal; Ogier, Simon; Ramon, Eloi; Nafria, Montserrat

被引用次数: 0

2024

Oxide Breakdown Spot Spatial Patterns as Fingerprints for Optical Physical Unclonable Functions

IEEE ELECTRON DEVICE LETTERS

Porti, Marc; Redon, Miquel; Munoz, Jordi; Nafria, Montserrat; Miranda, Enrique

被引用次数: 0

2023

Beneficial Role of Noise in Hf-based Memristors

2022 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS (ISCAS 22)

Rodriguez, Rosana; Martin-Martinez, Javier; Salvador, Emili; Crespo-Yepes, Albert; Miranda, Enrique; Nafria, Montserrat; Rubio, Antonio; Ntinis, Vasileios; Sirakoulis, Georgios Ch.

被引用次数: 4

2022

On the Impact of the Biasing History on the Characterization of Random Telegraph Noise

IEEE TRANSACTIONS ON INSTRUMENTATION AND MEASUREMENT

Saraza-Canflanca, Pablo; Castro-Lopez, Rafael; Roca, Elisenda; Martin-Martinez, Javier; Rodriguez, Rosana; Nafria, Montserrat; Fernandez, Francisco, V

被引用次数: 0

2022

Aging in CMOS RF Linear Power Amplifiers: An Experimental Study

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Aragones, Xavier; Barajas, Enrique; Crespo-Yepes, Albert; Mateo, Diego; Rodriguez, Rosana; Martin-Martinez, Javier; Nafria, Montserrat

被引用次数: 13

2021

Power-Efficient Noise-Induced Reduction of ReRAM Cell's Temporal Variability Effects

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Ntinis, Vasileios; Rubio, Antonio; Sirakoulis, Georgios Ch; Aguilera, Emili Salvador; Pedro, Marta; Crespo-Yepes, Albert; Martin-Martinez, Javier; Rodriguez, Rosana; Nafria, Montserrat

被引用次数: 16

2021

Statistical threshold voltage shifts caused by BTI and HCI at nominal and accelerated conditions

SOLID-STATE ELECTRONICS

Diaz-Fortuny, Javier; Saraza-Canflanca, Pablo; Rodriguez, Rosana; Martin-Martinez, Javier; Castro-Lopez, Rafael; Roca, Elisenda; V. Fernandez, Francisco; Nafria, Montserrat

被引用次数: 4

2021

Exploring the resistance change per energy unit as universal performance parameter for resistive switching devices

SOLID-STATE ELECTRONICS

Gomez, Jorge; Vourkas, Ioannis; Abusleme, Angel; Rodriguez, Rosana; Martin-Martinez, Javier; Nafria, Montserrat; Rubio, Antonio

被引用次数: 7

2020

Flexible Setup for the Measurement of CMOS Time-Dependent Variability With Array-Based Integrated Circuits

IEEE TRANSACTIONS ON INSTRUMENTATION AND MEASUREMENT

Diaz-Fortuny, Javier; Saraza-Canflanca, Pablo; Castro-Lopez, Rafael; Roca, Elisenda; Martin-Martinez, Javier; Rodriguez, Rosana; Fernandez, Francisco V.; Nafria, Montserrat

被引用次数: 16

2020

DC Characterization and Fast Small-Signal Parameter Extraction of Organic Thin Film Transistors With Different Geometries

IEEE ELECTRON DEVICE LETTERS

Arnal, August; Crespo-Yepes, Albert; Ramon, Eloi; Teres, Lluís; Rodriguez, Rosana; Nafria, Montserrat

被引用次数: 5

2020

A smart noise- and RTN-removal method for parameter extraction of CMOS aging compact models

SOLID-STATE ELECTRONICS

Diaz-Fortuny, Javier; Martin-Martinez, Javier; Rodriguez, Rosana; Castro-Lopez, Rafael; Roca, Elisenda; Fernandez, Francisco, V.; Nafria, Montserrat

被引用次数: 4

2019

Low-Power, High-Performance, Non-volatile Inkjet-Printed HfO₂-Based Resistive Random Access Memory: From Device to Nanoscale Characterization

ACS APPLIED MATERIALS & INTERFACES

Vescio, Giovanni; Martin, Gemma; Crespo-Yepes, Albert; Claramunt, Sergi; Alonso, Daniel; Lopez-Vidrier, Julian; Estrade, Sonia; Porti, Marc; Rodriguez, Rosana; Peiro, Francesca; Cornet, Albert; Cirera, Albert; Nafria, Montserrat

被引用次数: 18

2019

Self-Organizing Neural Networks Based on OxRAM Devices under a Fully Unsupervised Training Scheme

MATERIALS

Pedro, Marta; Martin-Martinez, Javier; Maestro-Izquierdo, Marcos; Rodriguez, Rosana; Nafria, Montserrat

被引用次数: 10

2019

Experimental Verification of Memristor-Based Material Implication NAND Operation

IEEE TRANSACTIONS ON EMERGING TOPICS IN COMPUTING

Maestro-Izquierdo, Marcos; Martin-Martinez, Javier; Crespo Yepes, Albert; Escudero, Manel; Rodriguez, Rosana; Nafria, Montserrat; Aymerich, Xavier; Rubio, Antonio

被引用次数: 3

2019

A Versatile CMOS Transistor Array IC for the Statistical Characterization of Time-Zero Variability, RTN, BTI, and HCI

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Diaz-Fortuny, Javier; Martin-Martinez, Javier; Rodriguez, Rosana; Castro-Lopez, Rafael; Roca, Elisenda; Aragonés, Xavier; Barajas, Enrique; Mateo, Diego; Fernandez, Francisco V.; Nafria, Montserrat

被引用次数: 34

2019

Performance and Power Consumption Trade-Off in UTBB FDSOI Inverters Operated at NTV for IoT Applications

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Couso, Carlos; Martin-Martinez, Javier; Porti, Marc; Nafria, Montserrat

被引用次数: 5

2018

Editorial to J-EDS ESSDERC 2017 Special Issue

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Nafria, Montserrat; Collaert, Nadine

被引用次数: 0

2018

Experimental Time Evolution Study of the HfO₂-Based IMPLY Gate Operation

IEEE TRANSACTIONS ON ELECTRON DEVICES

Maestro-Izquierdo, Marcos; Martin-Martinez, Javier; Yepes, Albert Crespo; Escudero, Manel; Rodriguez, Rosana; Nafria, Montserrat; Aymerich, Xavier; Rubio, Antonio

被引用次数: 6

2018

Protective nanometer films for reliable Cu-Cu connections

MICROELECTRONICS RELIABILITY

Berthold, Tobias; Benstetter, Guenther; Frammelsberger, Werner; Bogner, Manuel; Rodriguez, Rosana; Nafria, Montserrat

被引用次数: 0

2017

Evaluation of ultra-thin structures composed of graphene and high-k dielectrics for resistive switching memory applications

INTERNATIONAL JOURNAL OF NANOTECHNOLOGY

Wu, Qian; Claramunt, Sergi; Porti, Marc; Nafria, Montserrat; Aymerich, Xavier

被引用次数: 0

2016

Investigation of Filamentary Current Fluctuations Features in the High-Resistance State of Ni/HfO₂-Based RRAM

IEEE TRANSACTIONS ON ELECTRON DEVICES

Bargallo Gonzalez, Mireia; Martin-Martinez, Javier; Maestro, Marcos; Cruz Acero, Maria; Nafria, Montserrat; Campabadal, Francesca

被引用次数: 25

2016

Nanoscale characterization of copper oxide films by Kelvin Probe Force Microscopy

THIN SOLID FILMS

Berthold, Tobias; Benstetter, Guenther; Frammelsberger, Werner; Rodriguez, Rosana; Nafria, Montserrat

被引用次数: 11

2015

Nanoscale characterization of CH₃-terminated Self-Assembled Monolayer on copper by advanced scanning probe microscopy techniques

APPLIED SURFACE SCIENCE

Berthold, Tobias; Benstetter, Guenther; Frammelsberger, Werner; Rodriguez, Rosana; Nafria, Montserrat

被引用次数: 4

2015

In Situ Demonstration of the Link Between Mechanical Strength and Resistive Switching in Resistive Random-Access Memories

ADVANCED ELECTRONIC MATERIALS

Shi, Yuanyuan; Ji, Yanfeng; Hui, Fei; Nafria, Montserrat; Porti, Marc; Bersuker, Gennadi; Lanza, Mario

被引用次数: 12

2015

Conductive-AFM topography and current maps simulator for the study of polycrystalline high-k dielectrics

JOURNAL OF VACUUM SCIENCE & TECHNOLOGY B

Couso, Carlos; Porti, Marc; Martin-Martinez, Javier; Iglesias, Vanessa; Nafria, Montserrat; Aymerich, Xavier

被引用次数: 2

2015

New Weighted Time Lag Method for the Analysis of Random Telegraph Signals

IEEE ELECTRON DEVICE LETTERS

Martin-Martinez, Javier; Diaz, Javier; Rodriguez, Rosana; Nafria, Montserrat; Aymerich, Xavier

被引用次数: 52

2014

Analysis of Factors in the Nanoscale Physical and Electrical Characterization of High-K Materials by Conductive Atomic Force Microscope

INTEGRATED FERROELECTRICS

Zhang, Kai; Lanza, Mario; Shen, Ziyong; Fu, Qiang; Hou, Shimin; Porti, Marc; Nafria, Montserrat

被引用次数: 1

2014

A Conductive AFM Nanoscale Analysis of NBTI and Channel Hot-Carrier Degradation in MOSFETs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Wu, Qian; Bayerl, Albin; Porti, Marc; Martin-Martinez, Javier; Lanza, Mario; Rodriguez, Rosana; Velayudhan, Vikas; Nafria, Montserrat; Aymerich, Xavier; Bargallo Gonzalez, Mireia; Simoen, Eddy

被引用次数: 6

2014

Leakage current through the poly-crystalline HfO₂: Trap densities at grains and grain boundaries

JOURNAL OF APPLIED PHYSICS

Pirrotta, Onofrio; Larcher, Luca; Lanza, Mario; Padovani, Andrea; Porti, Marc; Nafria, Montserrat; Bersuker, Gennadi

被引用次数: 84

2013

Nanoscale and device level electrical behavior of annealed ALD Hf-based gate oxide stacks grown with different precursors

MICROELECTRONICS RELIABILITY

Bayerl, Albin; Lanza, Mario; Aguilera, Lidia; Porti, Marc; Nafria, Montserrat; Aymerich, Xavier; de Gendt, Stefan

被引用次数: 8

2013

Electrical and mechanical performance of graphene sheets exposed to oxidative environments

NANO RESEARCH

Lanza, Mario; Wang, Yan; Gao, Teng; Bayerl, Albin; Porti, Marc; Nafria, Montserrat; Zhou, Yangbo; Jing, Guangyin; Zhang, Yanfeng; Liu, Zhongfan; Yu, Dapeng; Duan, Huiling

被引用次数: 37

2013

Circuit Design-Oriented Stochastic Piecewise Modeling of the Postbreakdown Gate Current in MOSFETs: Application to Ring Oscillators

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Martin-Martinez, Javier; Kaczer, Ben; Degraeve, Robin; Roussel, Philippe J.; Rodriguez, Rosana; Nafria, Montserrat; Aymerich, Xavier; Dierickx, B.; Groeseneken, Guido

被引用次数: 9

2012

Resistive switching in hafnium dioxide layers: Local phenomenon at grain boundaries

APPLIED PHYSICS LETTERS

Lanza, M.; Bersuker, G.; Porti, M.; Miranda, E.; Nafria, M.; Aymerich, X.

被引用次数: 158

2012

Grain boundaries as preferential sites for resistive switching in the HfO₂ resistive random access memory structures

APPLIED PHYSICS LETTERS

Lanza, M.; Zhang, K.; Porti, M.; Nafria, M.; Shen, Z. Y.; Liu, L. F.; Kang, J. F.; Gilmer, D.; Bersuker, G.

被引用次数: 184

2012

Injected Charge to Recovery as a Parameter to Characterize the Breakdown Reversibility of Ultrathin HfSiON Gate Dielectric

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Crespo-Yepes, Albert; Martin-Martinez, Javier; Rothschild, Aude; Rodriguez, Rosana; Nafria, Montserrat; Aymerich, Xavier

被引用次数: 3

2011

Gate Voltage Influence on the Channel Hot-Carrier Degradation of High-k-Based Devices

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Amat, Esteve; Kauerauf, Thomas; Degraeve, Robin; Rodriguez, Rosana; Nafria, Montserrat; Aymerich, Xavier; Groeseneken, Guido

被引用次数: 13

2011

Nanoscale and Device Level Gate Conduction Variability of High-k Dielectrics-Based Metal-Oxide-Semiconductor Structures

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Bayerl, Albin; Lanza, Mario; Porti, Marc; Nafria, Montserrat; Aymerich, Xavier; Campabadal, F.; Benstetter, Guenther

被引用次数: 11

2011

Metal oxide resistive memory switching mechanism based on conductive filament properties

JOURNAL OF APPLIED PHYSICS

Bersuker, G.; Gilmer, D. C.; Veksler, D.; Kirsch, P.; Vandelli, L.; Padovani, A.; Larcher, L.; McKenna, K.; Shluger, A.; Iglesias, V.; Porti, M.; Nafria, M.

被引用次数: 456

2011

Conductivity and Charge Trapping After Electrical Stress in Amorphous and Polycrystalline Al₂O₃-Based Devices Studied With AFM-Related Techniques

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Lanza, Mario; Porti, Marc; Nafria, Montserrat; Aymerich, Xavier; Benstetter, Guenther; Lodermeier, Edgar; Ranzinger, Heiko; Jaschke, Gert; Teichert, Steffen; Wilde, Lutz; Michalowski, Pawel Piotr

被引用次数: 28

2011

Grain boundary-driven leakage path formation in HfO₂ dielectrics

SOLID-STATE ELECTRONICS

Bersuker, G.; Yum, J.; Vandelli, L.; Padovani, A.; Larcher, L.; Iglesias, V.; Porti, M.; Nafria, M.; McKenna, K.; Shluger, A.; Kirsch, P.; Jammy, R.

被引用次数: 112

2011

Combined Nanoscale and Device-Level Degradation Analysis of SiO₂ Layers of MOS Nonvolatile Memory Devices

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Lanza, Mario; Porti, Marc; Nafria, Montserrat; Aymerich, Xavier; Sebastiani, Alessandro; Ghidini, Gabriella; Vedda, Anna; Fasoli, M.

被引用次数: 12

2009

Channel Hot-Carrier Degradation in Short-Channel Transistors With High-k/Metal Gate Stacks

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Amat, Esteve; Kauerauf, Thomas; Degraeve, Robin; De Keersgieter, An; Rodriguez, Rosana; Nafria, Montserrat; Aymerich, Xavier; Groeseneken, Guido

被引用次数: 44

2009

Competing Degradation Mechanisms in Short-Channel Transistors Under Channel Hot-Carrier Stress at Elevated Temperatures

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Amat, Esteve; Kauerauf, Thomas; Degraeve, Robin; Rodriguez, Rosana; Nafria, Montserrat; Aymerich, Xavier; Groeseneken, Guido

被引用次数: 40

2009

Time-Dependent Variability Related to BTI Effects in MOSFETs: Impact on CMOS Differential Amplifiers

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Martin-Martinez, Javier; Rodriguez, Rosana; Nafria, Montserrat; Aymerich, Xavier

被引用次数: 17

2009

Understanding and Optimization of Hot-Carrier Reliability in Germanium-on-Silicon pMOSFETs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Maji, Debabrata; Crupi, Felice; Amat, Esteve; Simoen, Eddy; De Jaeger, Brice; Brunco, David P.; Manoj, C. R.; Rao, V. Ramgopal; Magnone, Paolo; Giusi, Gino; Pace, Calogero; Pantisano, Luigi; Mitard, Jerome; Rodriguez, Rosana; Nafria, Montserrat

被引用次数: 16

2009

Channel-Hot-Carrier Degradation and Bias Temperature Instabilities in CMOS Inverters

IEEE TRANSACTIONS ON ELECTRON DEVICES

Martin-Martinez, Javier; Gerardin, Simone; Amat, Esteve; Rodriguez, Rosana; Nafria, Montserrat; Aymerich, Xavier; Paccagnella, Alessandro; Ghidini, Gabriella

被引用次数: 21

2009

Effects of the Localization of the Charge in Nanocrystal Memory Cells

IEEE TRANSACTIONS ON ELECTRON DEVICES

Gasperin, Alberto; Amat, Esteve; Porti, Marc; Martin-Martinez, Javier; Nafria, Montserrat; Aymerich, Xavier; Paccagnella, Alessandro

被引用次数: 6

2009

Peculiar characteristics of nanocrystal memory cells programming window

JOURNAL OF VACUUM SCIENCE & TECHNOLOGY B

Gasperin, Alberto; Amat, Esteve; Martin, Javier; Porti, Marc; Nafria, Montserrat; Paccagnella, Alessandro

被引用次数: 0

2009

Gate oxide, wear-out and breakdown effects on the performance of analog and digital circuits

IEEE TRANSACTIONS ON ELECTRON DEVICES

Fernandez, Raul; Martin-Martinez, J.; Rodriguez, R.; Nafria, Montserrat; Aymerich, Xavier H.

被引用次数: 38

2008

学术合作



Montserrat Nafria

巴塞罗那自治大学

合作次数: 49

Marc Porti

巴塞罗那自治大学

合作次数: 21

Javier Martin Martinez

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合作次数: 12

Albert Crespo Yepes

巴塞罗那自治大学

合作次数: 3

申自勇

北京大学

合作次数: 2

还有 10 个合作者